

D2: Reconciling the biomass measurements, and testing both N_0 back-calculations

Does the published *Pseudomonas* CUE result change under a better-supported N_0 ?

OxygenModel D-series

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Verdict

The published temperature result does not change. The absolute respiration and CUE values do — by a factor of about 57.

Two independent measurements of the cells in a vial — the recorded OD_{600} and the flow-cytometry count — agree with each other to a median factor of 1.26. The quantity the pipeline actually uses, `N_inoculation_cells_per_L`, agrees with neither: it is a median **66×** below the flow-cytometry count and **49×** below the OD-derived count. Because $R = K O_{2,ref}/N_0$, per-cell respiration is high by that factor.

The shared assumption of both back-calculations survives testing: flow-cytometry fold change matches $\exp(r \Delta t)$ at the oxygen-derived r to a median of **0.990**, with 74 of 75 replicates inside a factor of two. So the forward route's $\exp(r\delta)$ correction is the right *size* (median 3.86×); it is the starting value that is wrong. Backward over forward has median **57** (range 18–496), and all 74 retained replicates exceed 10. For scale, D1 measured the fit-window choice as worth at most a factor of 1.23 on R .

But the published *Pseudomonas* temperature conclusion is untouched. Growth T_{opt} (33.98 °C) and respiration T_{opt} (37.53 °C) are identical to eight significant figures under all three routes, because neither quantity contains N_0 . The CUE optimum moves from **32.44 °C** to **32.37 °C** — a shift of 0.07 °C — and stays 1.5–1.7 °C below the growth optimum, inside the stated 30–35 °C band. The paper's stated conclusion holds under every route tested (§4.4).

One correction to the brief. `06_main_figures.R` does *not* use the taxon-mean initial count. It reads `data/OD_r_FC_r.csv`, whose `FC_Initial` is per-replicate; the recomputation reproduces D1's `r_FC` to exactly 0. Using the taxon mean instead would *degrade* the published validation (R^2 0.968 → 0.879). The published choice is the better one.

1 What this report does, and what it does not

Per-cell respiration is $R = K O_{2,ref}/N_0$, where N_0 is the cell density at the start of the fitted oxygen trajectory. In a sealed, headspace-free vial that density cannot be measured without destroying the closed-system measurement, so it is back-calculated. This is a considered design choice, not a defect; the question is only which back-calculation the data support.

`config.R` sets `NO_BACKPROJECT <- TRUE`, giving

$$N_0^{\text{forward}} = N_{\text{inoc}} \cdot e^{r\delta} \quad \implies \quad \log R = \log K + \log O_{2,ref} - \log N_{\text{inoc}} - r\delta$$

so respiration is an explicit *decreasing* function of the fitted growth rate. The alternative uses the endpoint count from the same vial:

$$N_0^{\text{backward}} = N_{\text{end}} \cdot e^{-r(t_{\text{end}}-t_0)}$$

Both assume exponential growth — the forward route across the roughly 45 min stabilisation period the paper itself says the oxygen signal cannot be trusted in, the backward route only across the interval the oxygen model already assumes it over. In the *Candida* application of this method the equivalent term could only be **bounded**, because no endpoint biomass existed. Here it can be **measured**.

Nothing was changed. `NO_BACKPROJECT` remains TRUE, the QC thresholds, fit windows, exclusions and cell sizes are untouched. `10_temperature_cue.R` gained an `OXYMODEL_NO_ROUTE` switch on the same pattern as D1's `OXYMODEL_RESULTS`, defaulting to the published behaviour; the forward

Table 1: Every conversion assumption. The two that are ASSUMED rather than stated are flagged as such.

Quantity	Value	Source
FC fixation dilution (490 ul + 10 ul glutaraldehyde)	1.020408	paper
FC PBS dilution (50 ul into 450 ul PBS)	10.0	paper
FC stain dilution (2.5 ul SYBR into 500 ul)	1.0050	paper
FC total dilution factor	10.2551	product
CytoFLEX S output unit	events per uL of the MEASURED (diluted) sample	ASSUMED
OD600 = 1 -> cells/mL, central	8e+08	E. coli
OD600 = 1 -> cells/mL, low	5e+08	document
OD600 = 1 -> cells/mL, high	2e+09	document
Taxon-specific OD-to-cell factor	NONE AVAILABLE	no published
E. coli reference cell volume	0.3600 μm^3	0.5 x
Size adjustment	linear in biovolume	APPENDIX
Paper's stated OD at aliquoting	0.0005	paper

run reproduces `runs/D1_baseline` byte-identically. `results/`, `data/` and `runs/D1_baseline/` were verified unchanged by checksum (57 of 57 sums OK, 0 failed).

2 Part A — the three statements of cell density

2.1 The conversions, and every assumption in them

The flow-cytometry dilution chain, shown as arithmetic:

$$\underbrace{\frac{500}{490}}_{\text{glutaraldehyde}} \times \underbrace{\frac{500}{50}}_{\text{PBS}} \times \underbrace{\frac{502.5}{500}}_{\text{SYBR}} = 10.2551$$

so an original culture density is the CytoFLEX reading $\times 10.2551 \times 10^6$ cells L^{-1} , **on the assumption that the instrument reports events per μL of the measured, diluted sample**. The paper does not state the output unit; this is the instrument default and is the largest single assumption in Part A.

No taxon-specific OD_{600} -to-cell factor is published for these 15 environmental isolates. The generic E. coli reference, 8×10^6 cells mL^{-1} per OD unit (documented spread $5 \times 10^6 - 2 \times 10^7$), is used for all of them, and a range is carried rather than a point estimate. **The OD route needs no dilution-chain assumption at all**, which is what makes the agreement below load-bearing.

Table 2: Cell density at inoculation, cells per litre, taxon medians. The stated-OD column (4.00e+08 for every taxon by construction) and the median OD600 are omitted for width. The last three columns are the pairwise ratios.

Taxon	Measured OD	Flow cyt.	N_inoc	FC / N_inoc	OD / N_inoc	FC / OD
Acinetobacter	5.92e+08	9.13e+08	3.35e+07	35	17	1.62
Aeromonas_A	1.79e+09	1.94e+09	3.23e+07	67	54	1.11
Aeromonas_B	1.67e+09	1.42e+09	1.21e+07	107	154	0.82
Arthrobacter	8.32e+08	2.07e+09	3.32e+07	63	27	2.48
Bacillus	5.60e+08	1.23e+08	9.87e+05	160	486	0.26
Burkholderia	1.07e+09	2.80e+09	3.67e+07	73	28	2.65
Buttiauxella	1.33e+09	2.30e+09	3.96e+07	61	34	1.67
Chromobacterium	1.60e+09	1.37e+09	1.84e+07	75	97	0.82
Chryseobacterium	6.80e+08	2.68e+09	8.14e+07	32	8	3.89
Flavobacterium_A	2.24e+09	1.51e+09	2.44e+07	59	83	0.76
Flavobacterium_B	5.60e+08	9.84e+08	1.81e+07	51	26	1.76
Herbaspirillum	7.52e+08	3.22e+09	7.16e+07	45	11	4.28
Pseudomonas	3.55e+09	1.32e+09	2.58e+07	58	138	0.44
Serratia	1.07e+09	1.27e+09	1.17e+07	91	61	1.49
Yersinia	2.96e+09	1.04e+09	5.89e+06	197	680	0.34

Table 3: Pairwise ratios across all 75 replicates.

Ratio	n	Min	25th	Median	75th	Max	Fold spread
FC_over_Ninoc	75	21.17	49.05	66.03	92.28	363.09	17
FC_over_measuredOD	75	0.20	0.73	1.26	2.30	5.62	28
FC_over_statedOD	75	0.28	2.87	3.74	5.68	10.08	36
measuredOD_over_Ninoc	75	5.61	25.57	49.41	108.98	1080.43	193
measuredOD_over_statedOD	75	0.80	1.73	2.88	4.63	10.66	13
statedOD_over_Ninoc	75	3.69	10.27	16.26	26.42	630.66	171

2.2 The three-way table

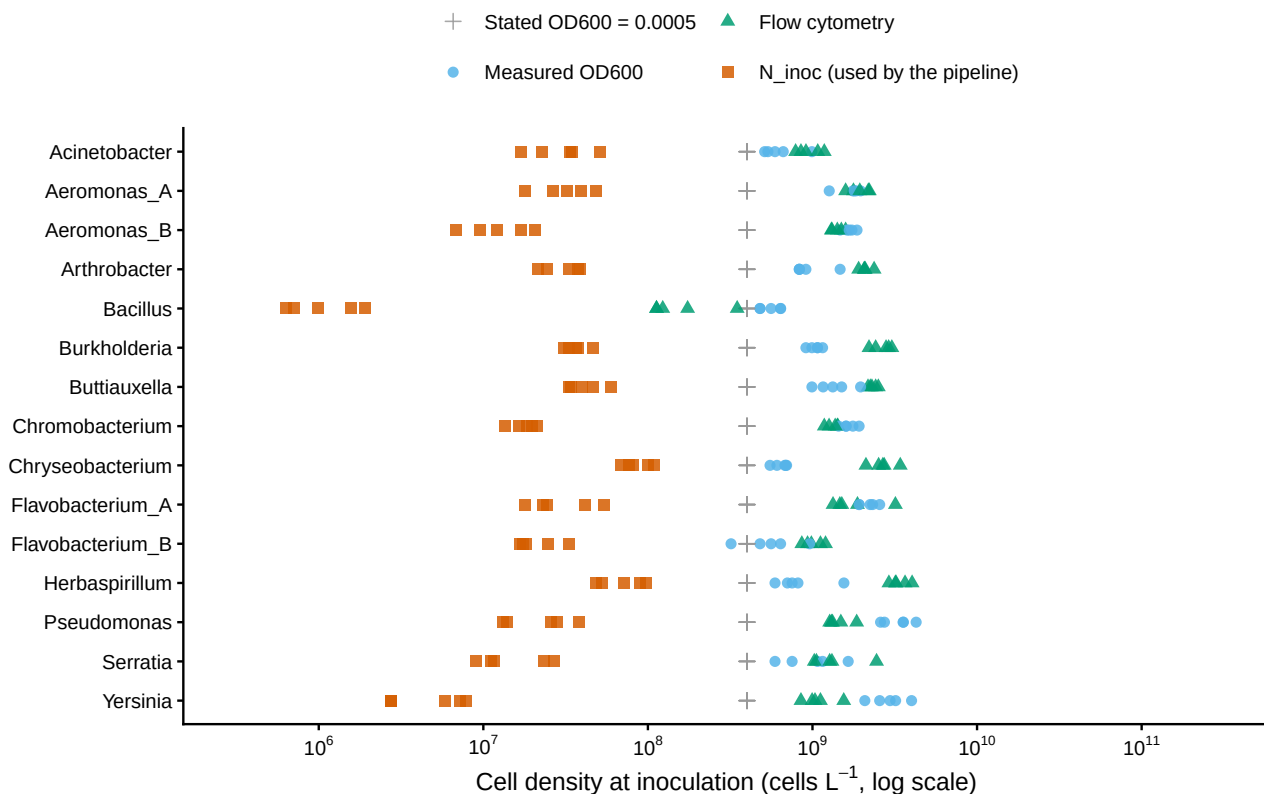


Figure 1: The three statements of cell density, per taxon and replicate. The two measurements overlap; the value the pipeline uses sits roughly two decades to their left.

Three things follow.

1. **The two measurements agree.** Flow cytometry over measured OD_{600} has median 1.26. Two entirely independent instruments, one of which needs no dilution assumption, land within about 26% of each other at the median.
2. **N_{inoc} agrees with neither**, and is low by a median $66\times$ against flow cytometry and $49\times$ against OD_{600} .
3. **The vials were not at the stated OD_{600} .** The recorded OD at aliquoting is a median $2.88\times$ the paper's stated 0.0005, ranging $0.80\text{--}10.66\times$.

2.3 Where does N_{inoc} come from? Not recovered.

The best candidate, the per-replicate $FC_{Initial}$, still leaves the implied factor spread over 17-fold (CV 0.55). $OD_{Initial}$ explains nothing ($\log\text{-log } r = 0.0004$). N_{inoc} is per-replicate, not per-taxon: all 15 taxa have 5 distinct values.

And it carries a signature no measurement produces. In 5 of the 15 taxa the largest N_{inoc} is exactly $3.000000\times$ the smallest, agreeing to 7–8 significant figures:

Table 4: If N_inoc were built from a candidate by a constant factor, the implied factor would be constant (CV = 0). Sorted by CV; none is close.

Candidate predictor	n	Implied factor (median)	CV of the factor	Fold spread	log-log r
FC_Initial (per replicate)	75	1.553e+05	0.546	17	0.879
FC_Initial (taxon mean)	75	1.514e+05	0.572	21	0.858
FC_Initial / cell_volume_um3	75	1.031e+05	0.584	25	0.854
constant (1)	75	2.461e+07	0.784	171	—
OD_Initial * FC_Initial	75	9.365e+07	0.928	79	0.608
OD_Initial / cell_carbon_fg	75	1.016e+12	1.037	169	0.125
OD_Initial / cell_volume_um3	75	1.016e+10	1.037	169	0.125
OD_Initial	75	1.619e+10	1.124	193	0.000
OD_Final	75	1.759e+09	1.234	578	-0.195
OD_Initial * cell_volume_um3	75	2.586e+10	1.294	345	-0.142
FC_Final (per replicate)	75	1.411e+04	1.302	380	0.457

Table 5: Taxa whose N_inoc range is exactly a factor of three.

Taxon	min N_inoc	max N_inoc	max / min
Acinetobacter	16,953,578	50,860,733	2.99999994
Aeromonas_B	6,844,556	20,533,669	3.00000015
Bacillus	634,257	1,902,770	2.99999842
Flavobacterium_A	17,999,458	53,998,373	2.99999994
Serratia	8,960,491	26,881,474	3.00000011

Taken together — no measured quantity reproduces it, and it has an exact arithmetic structure — **the provenance of N_inoc could not be recovered from the deposit.** That is a finding, not a failure of effort, and it is the first item in §6.

2.4 Cell_Counts.csv, and the parallel-vial reference density

Confirmed for 15 of 15 taxa. data/Cell_Counts.csv FC_Initial is constant across replicates and equals the rounded arithmetic mean of the per-replicate values in data/OD_r_FC_r.csv. The two files are otherwise identical:

Table 6: data/Cell_Counts.csv against data/OD_r_FC_r.csv, row by row.

Column	Rows	Identical	Differing
OD_Initial	75	75	0
OD_Final	75	75	0
FC_Initial	75	3	72
FC_Final	75	75	0

So Cell_Counts.csv FC_Initial is a **summary of the at-inoculation counts, not the parallel-vial 45-minute reference density the paper describes.** Searching the deposit for that parallel-vial

Table 7: Growth-rate agreement against the O2 method. Row 2 is what 06_main_figures.R actually computes, and it reproduces D1’s growth_rates_combined.csv to exactly 0.

Variant	n	Intercept	Slope	R2 (mixed)	R2 (OLS)	BA bias
r_O2 ~ r_OD600 (unchanged)	75	0.00287	0.9100	0.9435	0.7230	0.000859
r_O2 ~ r_FC, per-replicate initial (AS PUBL...	75	0.00310	0.8848	0.9680	0.9414	0.000482
r_O2 ~ r_FC, taxon-mean initial (COUNTERFAC...	75	0.00425	0.8418	0.8787	0.8595	0.000684

measurement finds nothing: no file, column or timestamp corresponds to it. It exists in the methods and not in the data.

D1 separately established that `data/Cell_Counts.csv` is read by no automated stage at all — only by the 04 Shiny app, as a fallback list of taxon names.

2.5 The effect of the taxon-mean-versus-per-replicate choice

The brief anticipated that the published validation used the taxon-mean initial, putting all replicate-level variation in the numerator. **It does not.** The counterfactual is worth reporting anyway, because it shows the published choice is the right one: switching to the taxon mean would drop R^2 from 0.968 to 0.879, move the slope from 0.885 to 0.842, and raise the Bland–Altman bias from 0.000482 to 0.000684 min^{-1} .

3 Part B — the two back-calculations

3.1 Test the shared assumption first

Both routes stand or fall on exponential growth at the oxygen-derived r over their interval. That is testable here, because the flow-cytometry fold change is an independent measurement of the same thing.

Table 8: $\text{FC_Final} / (\text{FC_Initial} \times \exp(r \times \text{duration}))$, all 75 replicates.

Statistic	Value
n replicates	75.0000
ratio min	0.4143
ratio 5th pct	0.6402
ratio 25th pct	0.8223
ratio median	0.9904
ratio 75th pct	1.0849
ratio 95th pct	1.2483
ratio max	1.7672
n within a factor of 2	74.0000
n within a factor of 1.5	69.0000
n excluded (primary, factor 2)	1.0000
n excluded (strict, factor 1.5)	6.0000
median r_FC / r_O2	0.9957

The assumption holds. The ratio has median 0.990, IQR 0.822–1.085, and 74 of 75 replicates sit within a factor of two. The median r_{FC}/r_{O_2} is 0.996. This is also an independent validation of the oxygen-derived growth rate itself.

One replicate is excluded from the backward route, with the reason recorded:

Table 9: Excluded from the backward route (kept if the ratio is within a factor of 2).

Taxon	Replicate	Observed fold	Predicted fold	Ratio	Reason
Flavobacterium_B	R4	2.74	6.60	0.414	growth saturated or cells died

A stricter factor-of-1.5 window would exclude 6 replicates instead of 1; the conclusions below are unchanged either way.

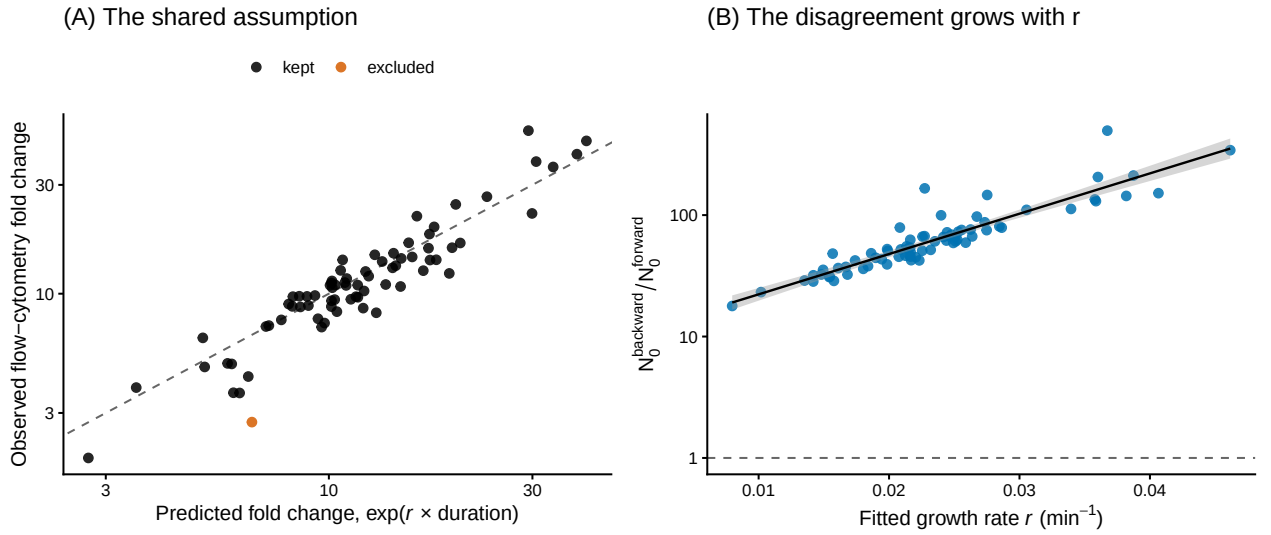


Figure 2: (A) The shared assumption: observed against predicted fold change, log-log, with the 1:1 line. (B) The disagreement between the two routes against the fitted growth rate.

3.2 Getting t_{end} right

The deposit gives two irreconcilable readings, and this matters — they differ by almost an order of magnitude in N_0 :

- **End of run.** The paper says endpoint samples were taken “from the corresponding SensorDish vials at the end of the run”. The oxygen runs end at 197–212 min. The endpoint sample time itself is not recorded.
- **The recorded interval.** `data/OD_r_FC_r.csv` carries a `Time` column of 101–139 min that `06_main_figures.R` uses as the initial-to-final interval.

They were tie-broken by a test that uses no endpoint at all: project the **measured initial** count forward to the fit-window start, $N_0 = N_{\text{start}} e^{rt_0}$, and ask which backward estimate agrees.

Table 11: Both N0 routes, all replicates.

Quantity	Value
n replicates (all)	75.000
n kept for the backward route	74.000
N0_forward median (cells/L)	9.274e+07
N0_backward median (cells/L), t_end = recorded interval	7.004e+09
N0_backward_alt median (cells/L), t_end = last oxygen reading	5.913e+08
N0_from_FCstart median (cells/L), no endpoint used	6.929e+09
CHECK backward / from_FCstart (median, should be 1)	0.990
CHECK backward_alt / from_FCstart (median, should be 1)	0.103
ratio backward/forward min	17.845
ratio backward/forward 25th pct	42.726
ratio backward/forward MEDIAN	57.134
ratio backward/forward 75th pct	75.903
ratio backward/forward max	496.168
ratio fold spread	27.805
n with ratio > 1	74.000
n with ratio > 10	74.000
forward amplification $\exp(r \cdot \delta)$ median	3.857
backward deflation $\exp(-r \cdot (t_{\text{end}} - t_0))$ median	0.331
ratio with t_end = last oxygen reading (median, SENSITIVITY)	6.696

Table 10: Tie-breaking t_end. A value of 1 means the backward estimate agrees with a projection that uses no endpoint at all.

Reading of t_end	Median agreement with the FC-start projection
the recorded OD/FC interval (101-139 min)	0.990
the end of the oxygen run (197-212 min)	0.103

The recorded interval is therefore used as the primary t_{end} and end-of-run is reported as the sensitivity. **Sensitivity:** using end-of-run instead would give a backward/forward ratio of 6.7 rather than 57 — still far from 1, so the direction of the finding does not depend on this choice, only its size.

3.3 Both estimates, and their ratio

Backward over forward has median **57**, IQR 43–76, range 17.8–496. All 74 retained replicates exceed 10, and none is near 1.

What this measures, rather than bounds. The forward route’s $\exp(r\delta)$ correction has median 3.86, and Part B’s growth check says growth over δ really is exponential at that rate. So the correction is the right *size*. The gap is almost entirely in the starting value: N_inoc is the median 66× below the measured count found in Part A, and $66 \times e^{rt_0}/e^{r\delta} \approx 57$. **The forward correction is not too large or too small; the number it is applied to is wrong.**

Table 12: Backward over forward, by taxon, ordered by the median ratio. Sorting by ratio also sorts by growth rate.

Taxon	n	N0 forward	N0 backward	Ratio min	Ratio median	Ratio max	r (min ⁻¹)
Flavobacterium_B	4	4.89e+07	1.33e+09	17.8	30.4	36.6	0.0145
Chryseobacterium	5	2.54e+08	8.68e+09	28.7	31.2	37.5	0.0158
Acinetobacter	5	7.01e+07	2.13e+09	23.2	31.9	43.3	0.0142
Herbaspirillum	5	2.81e+08	1.22e+10	38.0	42.4	50.9	0.0217
Buttiauxella	5	1.44e+08	7.13e+09	45.5	49.7	72.1	0.0216
Aeromonas_A	5	1.04e+08	5.37e+09	42.3	50.7	66.6	0.0215
Burkholderia	5	1.42e+08	7.58e+09	45.0	52.2	66.5	0.0209
Flavobacterium_A	5	1.06e+08	5.84e+09	36.1	55.3	66.0	0.0213
Arthrobacter	5	1.32e+08	8.05e+09	53.6	60.8	99.6	0.0244
Chromobacterium	5	9.14e+07	5.72e+09	51.7	62.9	76.1	0.0252
Pseudomonas	5	1.00e+08	5.94e+09	44.5	67.1	73.0	0.0227
Serratia	5	5.44e+07	6.31e+09	75.2	79.0	146.4	0.0275
Bacillus	5	5.49e+06	6.97e+08	66.8	97.2	166.0	0.0267
Aeromonas_B	5	7.87e+07	8.15e+09	78.8	112.4	134.6	0.0339
Yersinia	5	4.87e+07	1.05e+10	143.8	211.7	496.2	0.0382

3.4 And the disagreement is not a constant

Table 13: log(backward / forward) against the fitted growth rate.

Model	Intercept	Slope	SE	p	R2	Spearman rho
log(ratio) ~ r	2.338	76.45	4.003	6.67e-30	0.835	0.917
log(ratio) ~ log(r)	10.667	1.72	0.103	2.30e-26	0.793	–

log(ratio) against r has slope 76.5 min ($p = 6.67\text{e-}30$, $R^2 = 0.835$, Spearman $\rho = 0.917$). Fast-growing taxa disagree by 100–200 $\times$, slow-growing ones by about 30 $\times$. **Correcting N_0 would therefore change both the level of R and its dependence on r** — and it is that dependence that sets between-taxon respiration activation energies. This is the method-level finding, not a quirk of this dataset.

3.5 What it does to R and CUE

Set against D1’s sensitivity: the **fit-window choice moves R by at most a factor of 1.23; the N_0 route moves it by 57**. The two are not comparable, and the window sensitivity is not the binding uncertainty in this analysis.

4 Part C — the published temperature result under three routes

The temperature experiment has **no endpoint biomass** — data/Oxygen_Data_Filtered_CUE.csv carries only Taxon, Temperature, Replicate, Time, Oxygen — so a true backward route cannot be computed from it. *Pseudomonas* appears in both datasets, so the backward route transfers that taxon’s measured backward/forward ratio, 67.06. The third route, $N_0 = N_{\text{inoc}}$ with no back-calculation at all, is retained as a **lower bound and labelled as such, not as a candidate**.

Growth and respiration T_{opt} do not move at all — the shifts are 3.1e-06 and 3.6e-06 $^{\circ}\text{C}$, i.e. numerical noise. This is structural rather than lucky: the published parameter table fits growth T_{opt} to growth_r_per_hr, the raw growth rate, which contains no N_0 ; and respiration in 10_temperature_cue.R is $K \cdot O_2^{\text{ref}}$ in carbon units, which never contained N_0 either. **Only CUE can**

Table 14: Per-cell respiration R and growth G in fg C per cell per hour, and CUE, under each route. G does not depend on N0.

Taxon	n	R forward	R backward	R ratio	G	CUE fwd	CUE bwd
Flavobacterium_B	4	5301	187.0	0.0330	45.8	0.008	0.204
Chryseobacterium	5	415	13.3	0.0321	47.7	0.101	0.784
Acinetobacter	5	4658	157.2	0.0313	73.3	0.015	0.329
Herbaspirillum	5	408	10.2	0.0236	94.3	0.177	0.901
Buttiauxella	5	1452	27.2	0.0201	94.1	0.064	0.776
Aeromonas_A	5	1547	32.9	0.0197	87.1	0.053	0.726
Burkholderia	5	1501	28.0	0.0192	90.9	0.057	0.763
Flavobacterium_A	5	1965	35.5	0.0181	67.4	0.033	0.655
Arthrobacter	5	1646	27.1	0.0165	115.5	0.067	0.815
Chromobacterium	5	1968	29.6	0.0159	111.2	0.056	0.788
Pseudomonas	5	1765	29.3	0.0149	92.0	0.048	0.771
Serratia	5	2152	26.9	0.0127	84.3	0.037	0.746
Bacillus	5	31023	281.0	0.0103	177.1	0.006	0.387
Aeromonas_B	5	1346	11.0	0.0089	137.4	0.096	0.926
Yersinia	5	1637	6.1	0.0047	135.6	0.091	0.957

Table 15: Every effect on R that D1 and D2 have measured, on one scale.

Effect	Fold change in R	Direction
fit-window choice (D1: max relative difference in R)	1.23	either
forward back-projection $\exp(r \cdot \delta)$ alone (median)	3.86	lowers R
N_inoc vs the measured OD600 count (median, PART A)	49.41	lowers R
N_inoc vs the flow-cytometry count (median, PART A)	66.03	lowers R
N0 route: forward $\rightarrow$ backward, effect on R (median)	57.07	lowers R
N0 route: forward $\rightarrow$ backward, effect on R (max)	496.17	lowers R

Table 16: All optima, in degrees C, under the three N0 routes, against D1's published-route baseline.

Route	Growth Topt	Respiration Topt	CUE Topt	CUE - growth	shift vs D1 (growth)	shift vs D1 (resp)
forward	33.98	37.53	32.44	-1.54	3.1e-06	3.6e-06
backward	33.98	37.53	32.37	-1.61	3.1e-06	3.6e-06
none	33.98	37.53	32.31	-1.67	3.1e-06	3.6e-06

Table 17: The Sharpe-Schoolfield / Arrhenius parameters written by the pipeline under each route.

Route	Response	Best model	Topt	E (eV)	E 95% CI	Eh (eV)
forward	growth_r_per_hr	one	33.98	0.633	[0.571, 0.694]	4.872
forward	resp_C_fg_per_hr	one	37.53	0.712	[0.589, 0.836]	6.000
forward	carbon_use_efficiency	two	32.44	0.180	[-1.289, 1.649]	6.000
backward	growth_r_per_hr	one	33.98	0.633	[0.571, 0.694]	4.872
backward	resp_C_fg_per_hr	one	37.53	0.712	[0.589, 0.836]	6.000
backward	carbon_use_efficiency	two	32.37	0.178	[-1.292, 1.647]	6.000
none	growth_r_per_hr	one	33.98	0.633	[0.571, 0.694]	4.872
none	resp_C_fg_per_hr	one	37.53	0.712	[0.589, 0.836]	6.000
none	carbon_use_efficiency	one	32.31	0.100	[-0.135, 0.335]	6.000

Table 18: Per-temperature means, four replicates each, under all three routes.

Route	T (C)	N0 (cells/mL)	Growth (fg C/h)	Resp (fg C/h)	Resp:growth	CUE
forward	20	8.46e+05	3.28e+07	2.24e+08	6.82	0.1280
forward	24	1.02e+06	5.62e+07	2.41e+08	4.28	0.1904
forward	28	1.31e+06	1.02e+08	4.79e+08	4.77	0.1762
forward	32	1.64e+06	1.61e+08	6.15e+08	3.84	0.2091
forward	36	1.61e+06	1.55e+08	8.30e+08	5.38	0.1579
forward	40	7.85e+05	2.52e+07	6.84e+08	27.66	0.0355
backward	20	5.68e+07	2.20e+09	2.24e+08	0.10	0.9077
backward	24	6.82e+07	3.77e+09	2.41e+08	0.06	0.9400
backward	28	8.77e+07	6.84e+09	4.79e+08	0.07	0.9337
backward	32	1.10e+08	1.08e+10	6.15e+08	0.06	0.9459
backward	36	1.08e+08	1.04e+10	8.30e+08	0.08	0.9257
backward	40	5.26e+07	1.69e+09	6.84e+08	0.41	0.7090
none	20	5.50e+05	2.13e+07	2.24e+08	10.50	0.0871
none	24	5.50e+05	3.04e+07	2.41e+08	7.92	0.1129
none	28	5.50e+05	4.28e+07	4.79e+08	11.26	0.0824
none	32	5.50e+05	5.41e+07	6.15e+08	11.41	0.0814
none	36	5.50e+05	5.31e+07	8.30e+08	15.68	0.0602
none	40	5.50e+05	1.76e+07	6.84e+08	39.29	0.0250

Table 19: Does the paper’s stated conclusion survive each route?

Route	CUE at 20 C	Peak CUE	T of observed peak	CUE at 40 C	Increases	Declines at 40 C	Peaks in
forward	0.1280	0.2091	32	0.0355	yes	yes	yes
backward	0.9077	0.9459	32	0.7090	yes	yes	yes
none	0.0871	0.1129	24	0.0250	yes	yes	yes

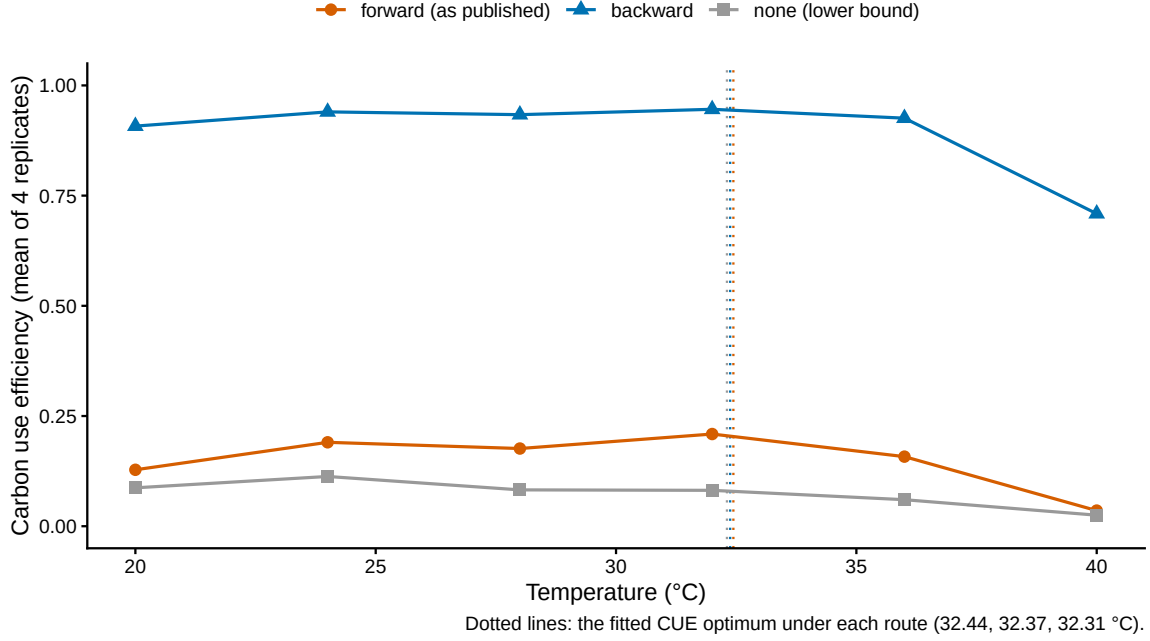


Figure 3: Carbon use efficiency against temperature under the three routes. The level moves by a factor of about 4.5; the optimum (dotted) moves by 0.13 degrees C.

What *does* change is the level: peak CUE is 0.209 (forward), 0.946 (backward) and 0.113 (lower bound), and the respiration-to-growth ratio at the optimum goes from 3.84 to 0.057. One honest nuance: under the lower bound the *model-free* per-temperature CUE peak moves from 32 °C to 24 °C even though the fitted optimum does not. Under the backward route — the actual candidate — both are unchanged.

Bottom line

No. The published *Pseudomonas* temperature result does not change materially under a better-supported N_0 . Growth T_{opt} stays at 33.98 °C and respiration T_{opt} at 37.53 °C — identical to eight significant figures, because neither quantity contains N_0 . The CUE optimum moves from 32.44 °C to 32.37 °C, a shift of 0.07 °C, and remains 1.61 °C below the growth optimum and inside the stated 30–35 °C band. CUE still rises with temperature, peaks at the growth optimum and falls at 40 °C under all three routes, including the lower bound. **The conclusion as stated in the paper is robust to the N_0 question.**

What is *not* robust is the absolute magnitude. Peak CUE is 0.21 as published and 0.95 under the backward route, and per-cell respiration across the bacterial set falls by a median factor of 57. Any statement about how much carbon a cell respire, or about the absolute value of CUE, rests on N_{inoc} — whose provenance could not be recovered, and which sits 49–66× below both independent measurements in the deposit. That is the part that warrants a conversation with the

authors, and it does not require a correction to the temperature figure.

5 Questions for the authors

Everything here is something the deposited data and the published methods could not settle.

1. **What is `N_inoculation_cells_per_L`, and how was it produced?** It matches no measured quantity in the deposit (best candidate leaves a 17-fold spread in the implied factor), sits 49–66× below both independent measurements, and in 5 of 15 taxa has a within-taxon range of exactly 3.000000. This is the single most consequential unknown: it scales every absolute respiration number in the paper.
2. **What unit does the CytoFLEX S output represent here** — events per μL of the diluted sample, or already dilution-corrected? The whole flow-cytometry arm of Part A rests on this, though the OD cross-check does not.
3. **Where are the parallel-vial 45-minute reference densities?** The methods describe them; no file, column or timestamp in the deposit corresponds to them. `Cell_Counts.csv` is *not* them — it is the rounded taxon mean of the at-inoculation counts (confirmed 15/15).
4. **When were the endpoint OD_{600} /FC samples actually taken?** The text says at the end of the run (197–212 min); `OD_r_FC_r.csv` carries an interval of 101–139 min. The two give N_0 estimates a factor of about 10 apart.
5. **Why is the recorded OD_{600} at aliquoting a median $2.9\times$ the stated 0.0005**, ranging up to $10.7\times$?
6. **Is there a taxon-specific OD_{600} -to-cells calibration** for these 15 isolates? Part A had to use a generic *E. coli* factor for all of them.
7. **Was the deposited `Oxygen_r` in `OD_r_FC_r.csv` produced by the pipeline in this repository?** It differs from both the modular and the original-script growth rates (e.g. *Acinetobacter* R1: 0.019508 in the file, 0.019436 modular, 0.019498 original).

Which findings are method-level

Method-level — these apply to the published analysis *and* to the *Candida* application of the same method, and to any future dataset:

- The forward route makes $\log R$ an explicit decreasing function of the fitted r . Whenever N_{inoc} is uncertain, that uncertainty enters R *correlated with* r , not as a constant offset (§3.5: slope 76.5 min, $R^2 = 0.835$). Between-taxon orderings of respiration activation energy are therefore not robust to it.
- Growth and respiration thermal optima, as this pipeline computes them, are *structurally* independent of N_0 . Only CUE can move. Any N_0 debate is a debate about CUE levels and about R , never about T_{opt} for growth or respiration.
- Where an endpoint biomass measurement exists, the back-projection can be **measured** rather than bounded — and the measurement here says the correction’s *form* was right and its *anchor* was not.

Dataset-specific — these are about this deposit only:

- The `N_inoc` provenance gap and its exact-factor-of-three signature.
- The missing parallel-vial reference densities.
- The t_{end} ambiguity.
- The recorded OD_{600} being about $2.9\times$ the stated value.

Table 21: Number-to-source map.

Section	Source
§2.1 conversion assumptions	runs/D2_analysis/A2_conversion_assumptions.csv
§2.2 three-way table	runs/D2_analysis/A2_three_way_density_by_taxon.csv, A2_three_way_den
§2.2 pairwise ratios	runs/D2_analysis/A2_pairwise_ratio_summary.csv
§2.3 N_inoc provenance	runs/D2_analysis/A3_Ninoc_provenance_candidates.csv, A3_Ninoc_structure
§2.4 Cell_Counts check	runs/D2_analysis/A1_CellCounts_is_taxon_mean.csv, A1_CellCounts_vs_O
§2.5 r_FC recomputation	runs/D2_analysis/A4_agreement_published_vs_perrep.csv
§3.1 growth check	runs/D2_analysis/B1_growth_check_summary.csv, B1_exponential_growth_c
§3.2 t_end tie-break	runs/D2_analysis/B2_N0_ratio_summary.csv
§3.3 N0 both routes	runs/D2_analysis/B2_N0_ratio_summary.csv, B2_N0_ratio_by_taxon.csv
§3.4 ratio against r	runs/D2_analysis/B2_N0_ratio_vs_r.csv
§3.5 R and CUE	runs/D2_analysis/B3_R_and_CUE_by_taxon.csv, B3_effect_sizes_vs_wind
§4 optima, three routes	runs/D2_analysis/C4_optima_three_routes.csv, C1_thermal_parameters_thr
§4 per temperature	runs/D2_analysis/C2_per_temperature_three_routes.csv
§4 Fig 7 panel A optimum	runs/D2_analysis/C3_growth_carbon_Topt_three_routes.csv
§4 conclusion test	runs/D2_analysis/C4_conclusion_holds_three_routes.csv
Pseudomonas transfer scale	runs/D2_analysis/C0_pseudomonas_scale.txt
D1 window sensitivity (1.23x)	runs/D1_baseline/comparisons/D2_per_quantity.csv
Figures 1-3	reports/D2_n0_routes/figures/, from d2_figures.R
Provenance	env/versions.json

Provenance

Table 20: Provenance. Environment fields come from env/versions.json.

Field	Value
Report	D2 — N0 reconciliation and dual route
Git commit (environment record)	0db2697a1da3841456a800eb26173edc89cd5912
Branch	main
Environment recorded (UTC)	2026-07-26 20:07:56 UTC
R	R version 4.5.2 (2025-10-31)
Platform	aarch64-apple-darwin20
OS	macOS Tahoe 26.2
renv lockfile	renv.lock — 135 packages, R 4.5.2
Quarto	1.8.27
PDF engine	lualatex — This is LuaHBTeX, Version 1.24.0 (TeX Liv...
Baseline compared against	runs/D1_baseline (D1, verified byte-identical)
Route output trees	runs/D2_forward, runs/D2_backward, runs/D2_none

Where every number came from

Artefacts are produced by `reports/D2_n0_routes/scripts/d2_partA_reconcile.R`, `d2_partB_routes.R`, `d2_partC_temperature.R` and `d2_figures.R`. The three route trees come

from `10_temperature_cue.R` run with `OXYMODEL_NO_ROUTE` set; the forward run reproduces `runs/D1_baseline` byte-identically, which is how the “no default was changed” claim is verified.

References